

Module 6. Cluster Analysis and Selected Methods

- ❑ Cluster Analysis: Basic Concepts 
- ❑ Partitioning Methods
- ❑ Hierarchical Methods
- ❑ Density-based and Grid-based Methods
- ❑ Advanced Methods
- ❑ Evaluation of Clustering
- ❑ Summary

Cluster Analysis: A Quick Overview

- ❑ When flying over a city, one can easily identify fields, forests, commercial areas, and residential areas based on their features, without anyone's explicit "training"—This is the power of cluster analysis
- ❑ This module systematically studies cluster analysis methods and helps answer the following:
 - ❑ What are the different proximity measures for effective clustering?
 - ❑ Can we cluster a massive number of data points efficiently?
 - ❑ Can we find clusters of arbitrary shape? At multiple levels of granularity?
 - ❑ How can we judge the quality of the clusters discovered by our system?

What Is Cluster Analysis?

- ❑ What is a cluster?
 - ❑ A cluster is a collection of data objects which are
 - ❑ Similar (or related) to one another within the same group (i.e., cluster)
 - ❑ Dissimilar (or unrelated) to the objects in other groups (i.e., clusters)
- ❑ Cluster analysis (or clustering, data segmentation, ...)
 - ❑ Given a set of data points, partition them into a set of groups (i.e., clusters) which are as similar as possible
- ❑ Cluster analysis is unsupervised learning (i.e., no predefined classes)
 - ❑ This contrasts with classification (i.e., supervised learning)
- ❑ Typical ways to use/apply cluster analysis
 - ❑ As a stand-alone tool to get insight into data distribution, or
 - ❑ As a preprocessing (or intermediate) step for other algorithms

The Value of Cluster Analysis

☐ What is the value of cluster analysis?

- ☐ Cluster analysis helps you partition massive data into groups based on its features
- ☐ Cluster analysis will often help subsequent data mining processes such as pattern discovery, classification, and outlier analysis

☐ What roles does cluster analysis play in the Data Mining Specialization?

- ☐ You will learn various scalable methods to find clusters from massive data
- ☐ You will learn how to mine different kinds of clusters effectively
- ☐ You will also learn how to evaluate the quality of the clusters you find
- ☐ Cluster analysis will help with classification, outlier analysis, and other data mining tasks